

7 Discrete Time Fourier Series

In continuous time we summed over infinite basis functions, but in discrete we have a limited number N .

We can reconstruct a function from Fourier coefficients with

$$x[n] = \sum_{k \in \langle N \rangle} a_k e^{jk\hat{\omega}_0 n}$$

Or find the coefficients as

$$a_k = \frac{1}{N} \sum_{n \in \langle N \rangle} x[n] e^{-jk\hat{\omega}_0 n}$$

As long as the range in the sum spans the full period. i.e. if $N=5$ then

$$\sum_0^4 \text{ or } \sum_{10}^{14} \text{ etc}$$

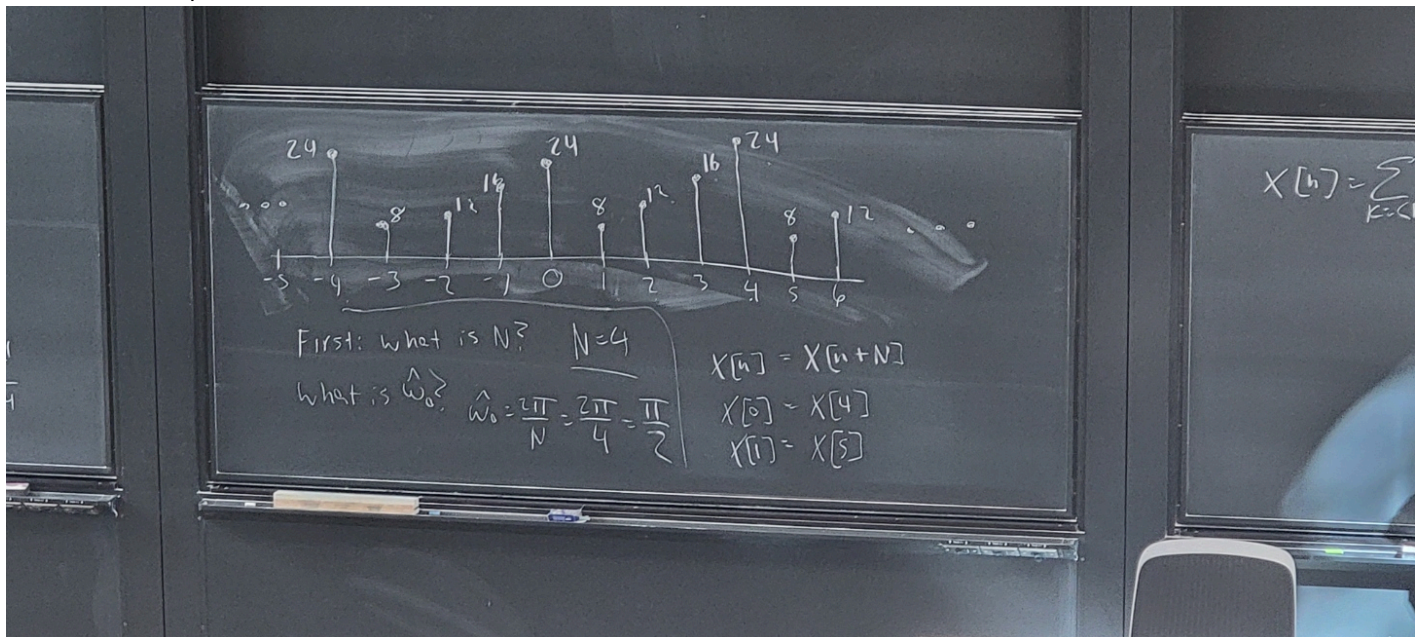
We define the period as N that satisfies

$$x[n] = x[n + N]$$

$$\hat{\omega}_0 = \frac{2\pi}{N}$$

Example 1

Lets do an example.



$$N = 4, \hat{\omega}_0 = \frac{\pi}{2}$$

Let's pick a naive range bc it doesn't matter

$$a_k = \frac{1}{4} \sum_{n=0}^3 x[n] e^{-jk\hat{\omega}_0 n}$$

We define

$$a_0 = \frac{1}{4} [x[0]e^{-j(0)\hat{\omega}_0 \underbrace{(0)}_k \underbrace{(0)}_n}] + \frac{1}{4} x[1]e^{-j(0)\hat{\omega}_0 \underbrace{(1)}_n} + \frac{1}{4} x[2]e^{-j(0)\hat{\omega}_0 \underbrace{(2)}_n} \dots$$

$$a_0 = \frac{1}{4}(24 + 8 + 12 + 16) = 15$$

Now we can find

$$a_1 = \frac{1}{4} [x[0]e^{-j(1)\hat{\omega}_0 \underbrace{(0)}_k \underbrace{(0)}_n}] + \frac{1}{4} x[1]e^{-j(1)\hat{\omega}_0 \underbrace{(1)}_n} + \frac{1}{4} x[2]e^{-j(1)\hat{\omega}_0 \underbrace{(2)}_n} \dots$$

$$a_1 = \frac{1}{4}(24 - 8j - 12 + 16j) = 3 + 2j$$

$$a_2 = \frac{1}{4} [x[0] + x[1]e^{-j\pi} + x[2]e^{-2\pi j} + x[3]e^{-3\pi j}]$$

$$= \frac{1}{4}(24 - 8 + 12 - 16) = 3$$

$$a_3 = \frac{1}{4} [x_0 + x[1]e^{-\frac{3\pi}{2}j} + x[2]e^{-3\pi j} + x[3]e^{-\frac{\pi}{2}j}]$$

$$= \frac{1}{4}[24 + 8j - 12 - 16j] = 3 - 2j$$

This gives us

$$\begin{bmatrix} a_0 & = & 15 \\ a_1 & = & 3 + 2j \\ a_2 & = & 3 \\ a_3 & = & 3 - 2j \end{bmatrix}$$

Because it doesn't depend on what starting indices, we have more Fourier coefficients (which don't contribute anything new)

$$a_k = a_{k+N}$$

$$a_k = 0 \text{ otherwise}$$

We can check that we'll recover our original signal

$$x[n] = \sum_{k=0}^3 a_k e^{jk\hat{\omega}_0 n}$$

$$= 15e^{0} + 3.6e^{0.5j}e^{\frac{\pi}{2}j} + 3e^{\pi nj} + 3.6e^{-0.5j}e^{\frac{3\pi}{2}nj}$$

We can simplify this with sin and cos

For example,

$$3e^{j\pi n} = \cos(\pi n) + j\sin(\pi n) \xrightarrow{0}$$

also,

$$e^{\frac{3\pi}{2}nj} = e^{j(\frac{4\pi}{2} - \frac{\pi}{2})n} = e^{2\pi nj}e^{-\frac{j\pi}{2}n} = e^{-\frac{\pi}{2}nj}$$

This just leaves the two gross terms

$$3.6e^{j(\frac{\pi}{2}n+0.5)} + 3.6e^{-j(\frac{\pi}{2}n+0.5)}$$

which just looks like the identity

$$\cos(\theta) = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$$

Example 2

Lets do another example

Lets take a simple thing

$$x[n] = \cos\left(\frac{\pi}{2}n\right)$$

$$N = 4$$

$$x[n] = \frac{1}{2}[e^{j\frac{\pi}{2}n} + e^{-j\frac{\pi}{2}n}]$$

We can solve by inspection,

$$a_1 = \frac{1}{2}, a_{-1} = \frac{1}{2}$$

Remember that the span is 4, so

$$\begin{aligned}a_1 &= a_5 = a \dots \\a_{-1} &= a_3 = a_{-5} \dots\end{aligned}$$